

CSE 4615:: Wireless Networks

Enhancing Insight into Wireless Transmission Fundamentals

Lecture 3

CHAPTER 2

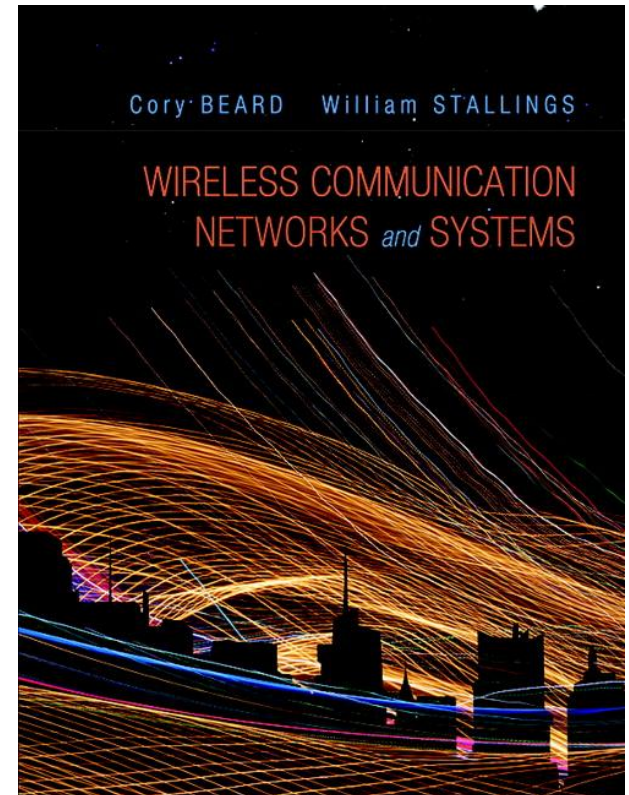
TRANSMISSION FUNDAMENTALS

These slides are made available to faculty in PowerPoint form. Slides can be freely added, modified, and deleted to suit student needs. They represent substantial work on the part of the authors; therefore, we request the following.

If these slides are used in a class setting or posted on an internal or external www site, please mention the source textbook and note our copyright of this material.

All material copyright 2016

Cory Beard and William Stallings, All Rights Reserved



Wireless Communication Networks and Systems

1st edition

Cory Beard, William Stallings

© 2016 Pearson Higher
Education, Inc.

CHAPTER 5 & 6 & 7

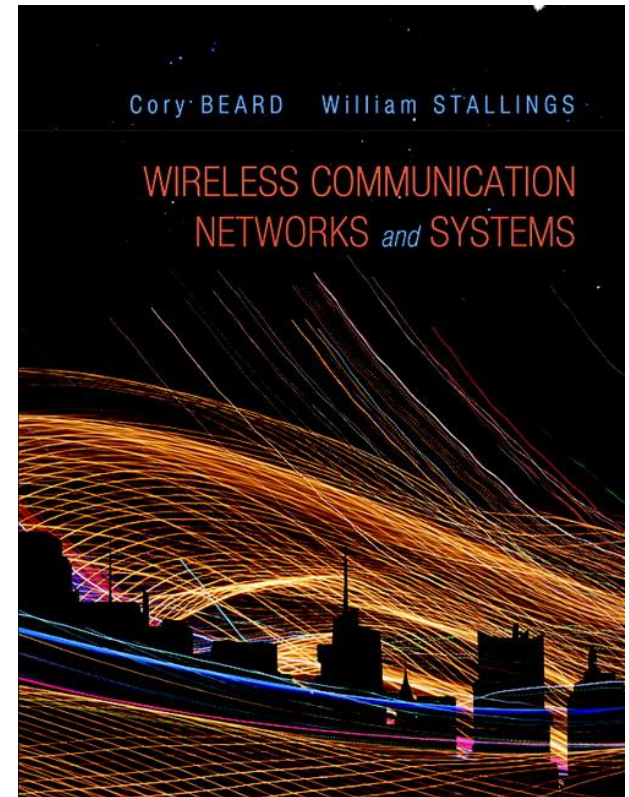
THE WIRELESS CHANNEL

These slides are made available to faculty in PowerPoint form. Slides can be freely added, modified, and deleted to suit student needs. They represent substantial work on the part of the authors; therefore, we request the following.

If these slides are used in a class setting or posted on an internal or external www site, please mention the source textbook and note our copyright of this material.

All material copyright 2016

Cory Beard and William Stallings, All Rights Reserved



Wireless Communication Networks and Systems

1st edition

Cory Beard, William Stallings

© 2016 Pearson Higher
Education, Inc.

CHAPTER 8

ORTHOGONAL

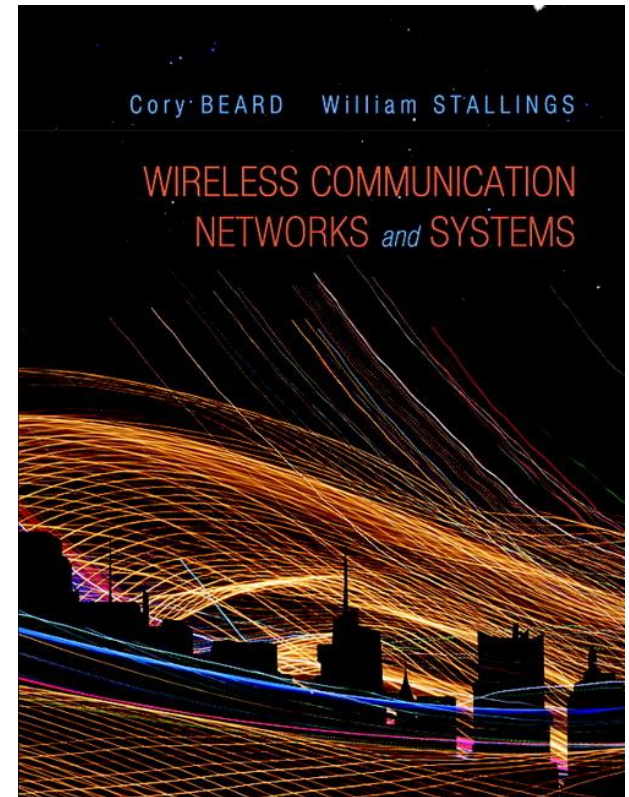
FREQUENCY DIVISION MULTIPLEXING

These slides are made available to faculty in PowerPoint form. Slides can be freely added, modified, and deleted to suit student needs. They represent substantial work on the part of the authors; therefore, we request the following.

If these slides are used in a class setting or posted on an internal or external www site, please mention the source textbook and note our copyright of this material.

All material copyright 2016

Cory Beard and William Stallings, All Rights Reserved



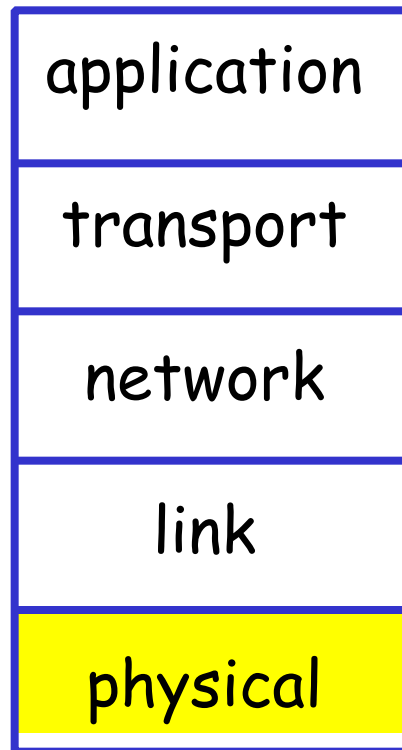
Wireless Communication Networks and Systems

1st edition

Cory Beard, William Stallings

© 2016 Pearson Higher
Education, Inc.

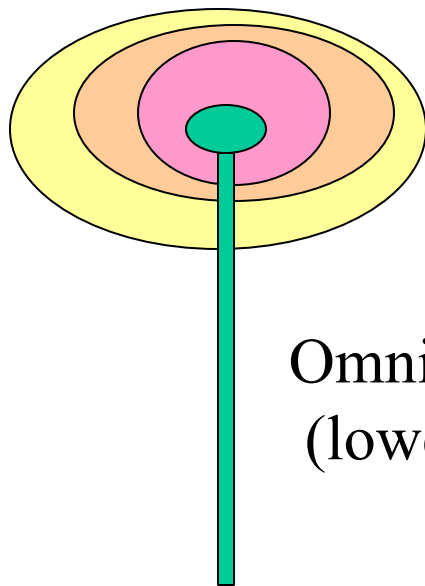
Physical Layer: Wireless Communication



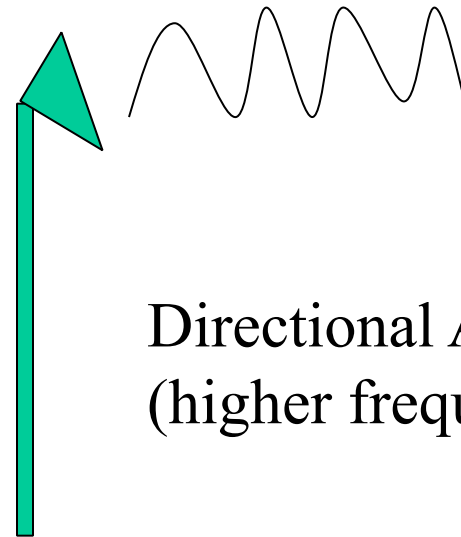
Antennas

Antennas

- An antenna is an electrical conductor or system of conductors to send/receive RF signals
 - Transmission - radiates electromagnetic energy into space
 - Reception - collects electromagnetic energy from space
- In two-way communication, the same antenna can be used for transmission and reception



Omnidirectional Antenna
(lower frequency)

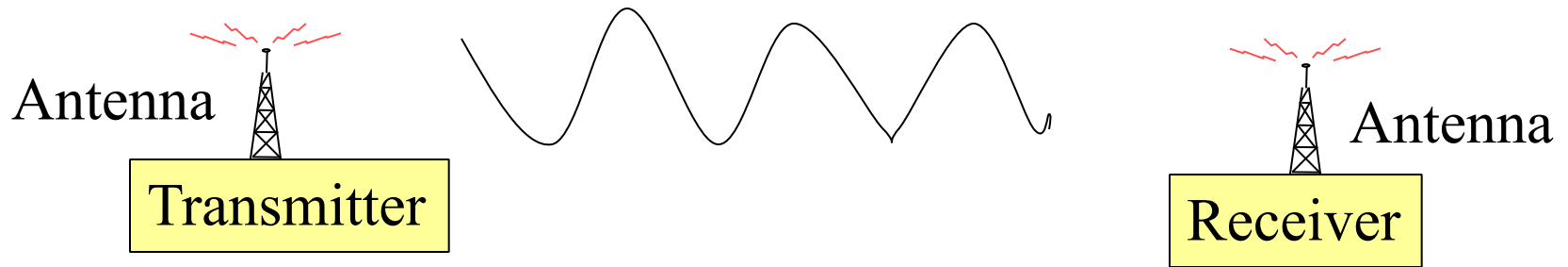


Directional Antenna
(higher frequency)

Antennas

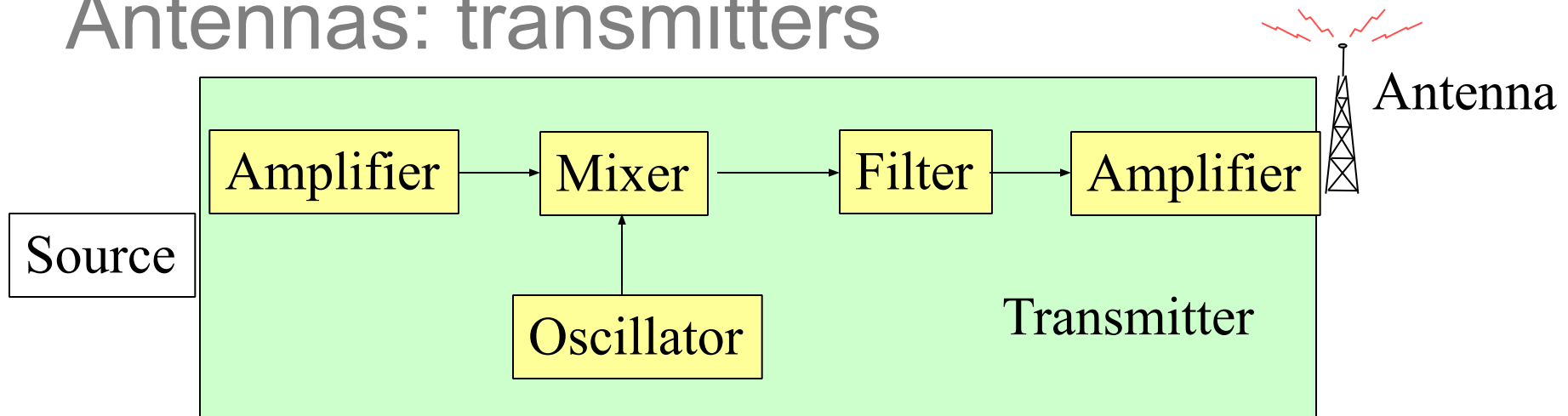
- Couples Electromagnetic energy to and from space and from a wire or coaxial cable.
- Real antennas have directive effect
- Directive antennas are used to overcome the environmental challenges
- Two or more antennas can be combine to improve reception called multi-element antenna array
- Antennas should be apart

Antennas: wireless transmission



- Wireless communication systems consist of:
 - Transmitters
 - Antennas: radiates electromagnetic energy into air
 - Receivers

Antennas: transmitters



Suppose you want to generate a signal that is sent at 900 MHz and the original source generates a signal at 300 MHz.

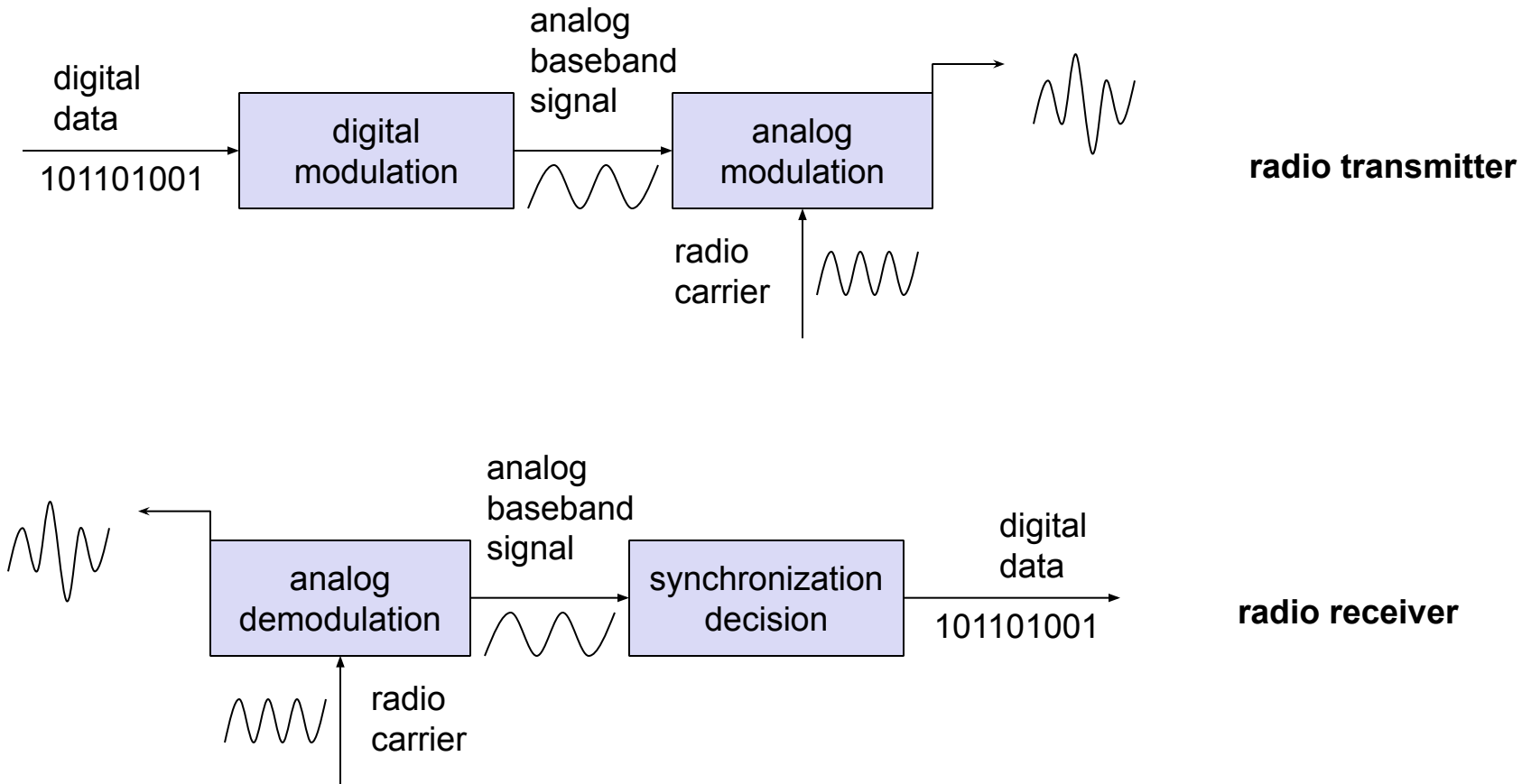
- Amplifier - strengthens the initial signal
- Oscillator - creates a carrier wave of 600 MHz
- Mixer - combines signal with oscillator and produces 900 MHz (also does modulation, etc)
- Filter - selects correct frequency
- Amplifier - Strengthens the signal before sending it

Modulation

Modulation

- What is modulation and types of modulation?
 - **Process of converting data into electrical signals optimized for transmission.**
- roughly divided into four types:
 - Analog modulation
 - Digital modulation
 - Pulse modulation
 - Spread spectrum method

Modulation and demodulation

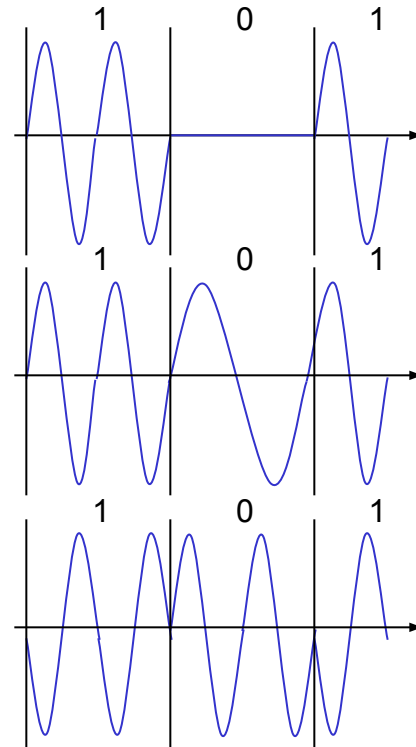


Modulation

- Digital modulation
 - digital data is translated into an analog signal (baseband)
 - ASK, FSK, PSK
 - differences in spectral efficiency, power efficiency, robustness
- Analog modulation
 - shifts center frequency of baseband signal up to the radio carrier
- Basic schemes
 - Amplitude Modulation (AM)
 - Frequency Modulation (FM)
 - Phase Modulation (PM)

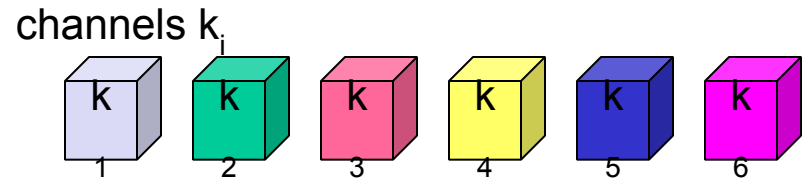
Digital modulation

- Modulation of digital signals known as Shift Keying
- Amplitude Shift Keying (ASK):
 - very simple
 - low bandwidth requirements
 - very susceptible to interference
- Frequency Shift Keying (FSK):
 - needs larger bandwidth
- Phase Shift Keying (PSK):
 - more complex
 - robust against interference



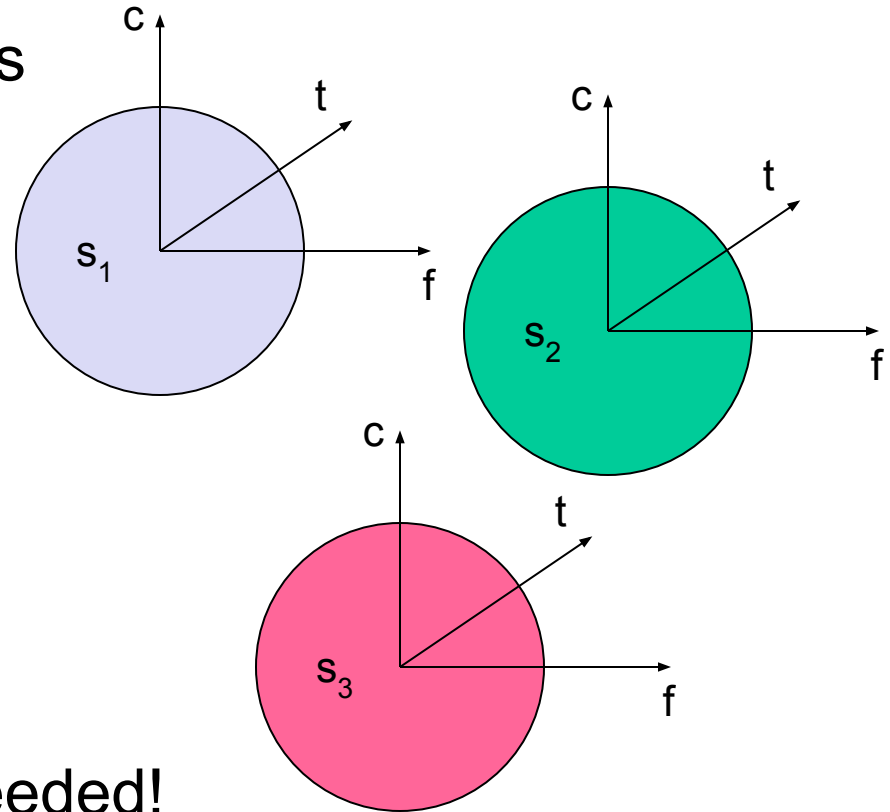
Multiplexing

Multiplexing



- Multiplexing in 4 dimensions

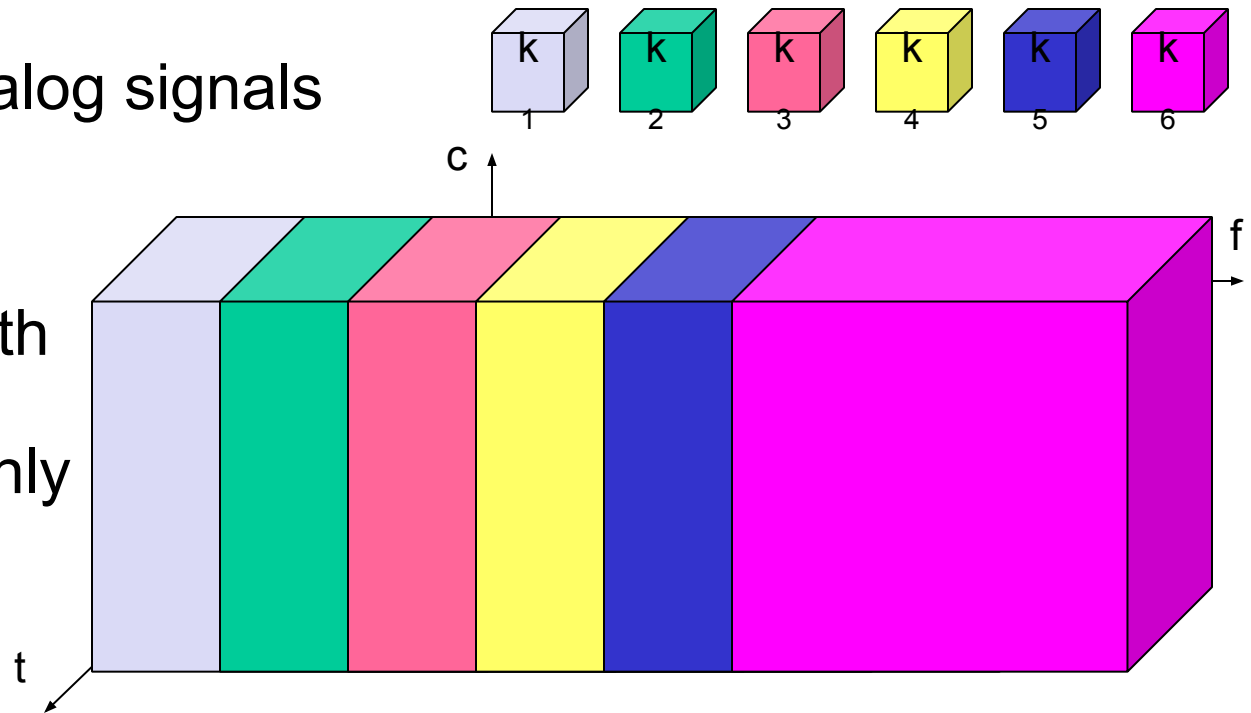
- space (s_i)
- time (t)
- frequency (f)
- code (c)



- Goal: multiple use of a shared medium
- Important: guard spaces needed!

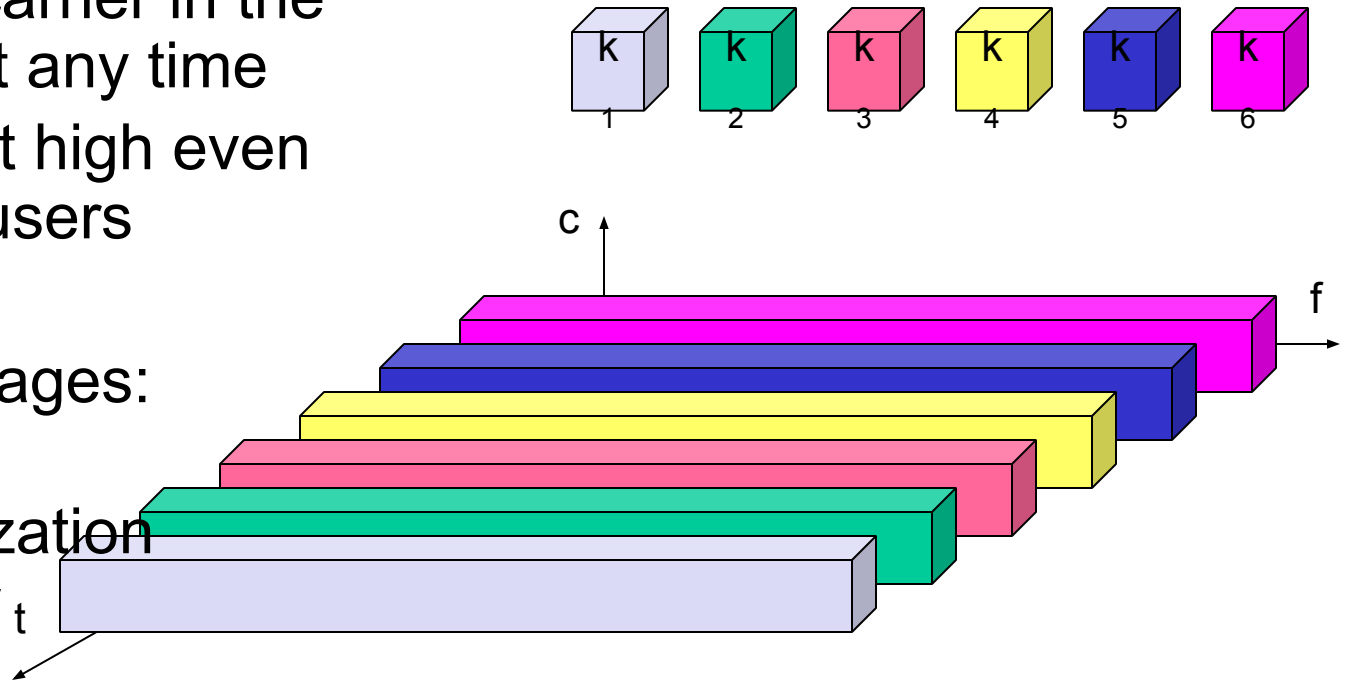
Frequency multiplex

- Separation of the whole spectrum into smaller frequency bands
- A channel gets a certain band of the spectrum for the whole time
- Advantages:
 - no dynamic coordination necessary
 - works also for analog signals
- Disadvantages:
 - waste of bandwidth if the traffic is distributed unevenly
 - inflexible
 - guard spaces



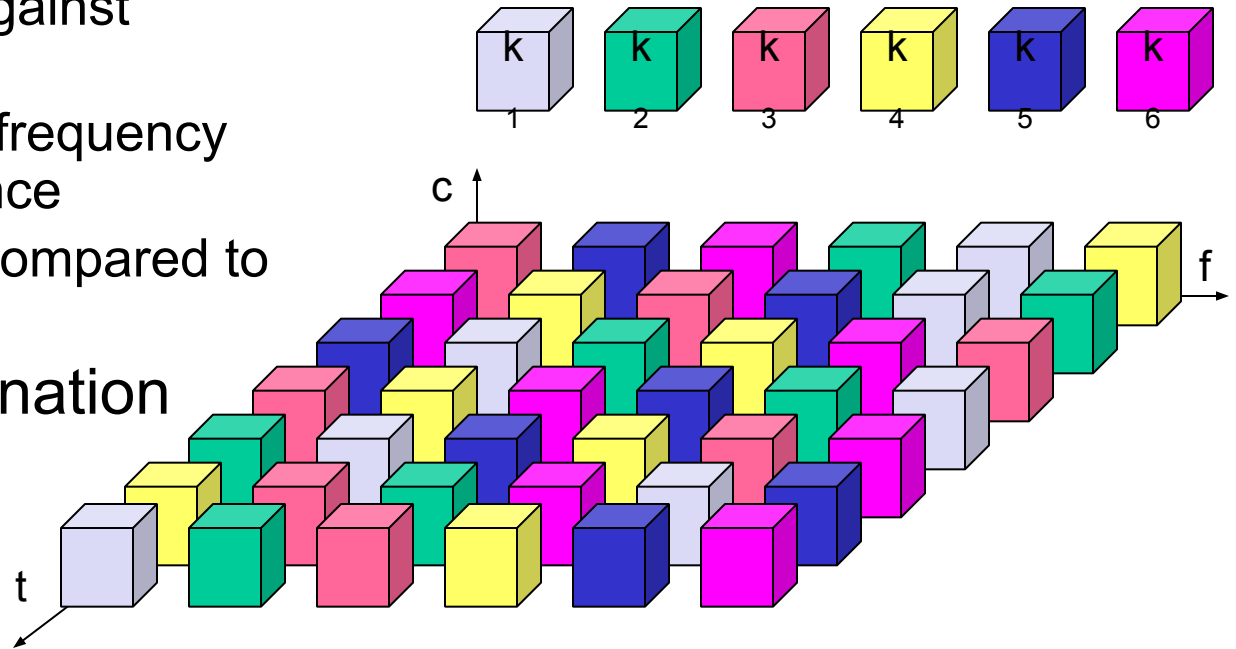
Time multiplex

- A channel gets the whole spectrum for a certain amount of time
- Advantages:
 - only one carrier in the medium at any time
 - throughput high even for many users
- Disadvantages:
 - precise synchronization necessary



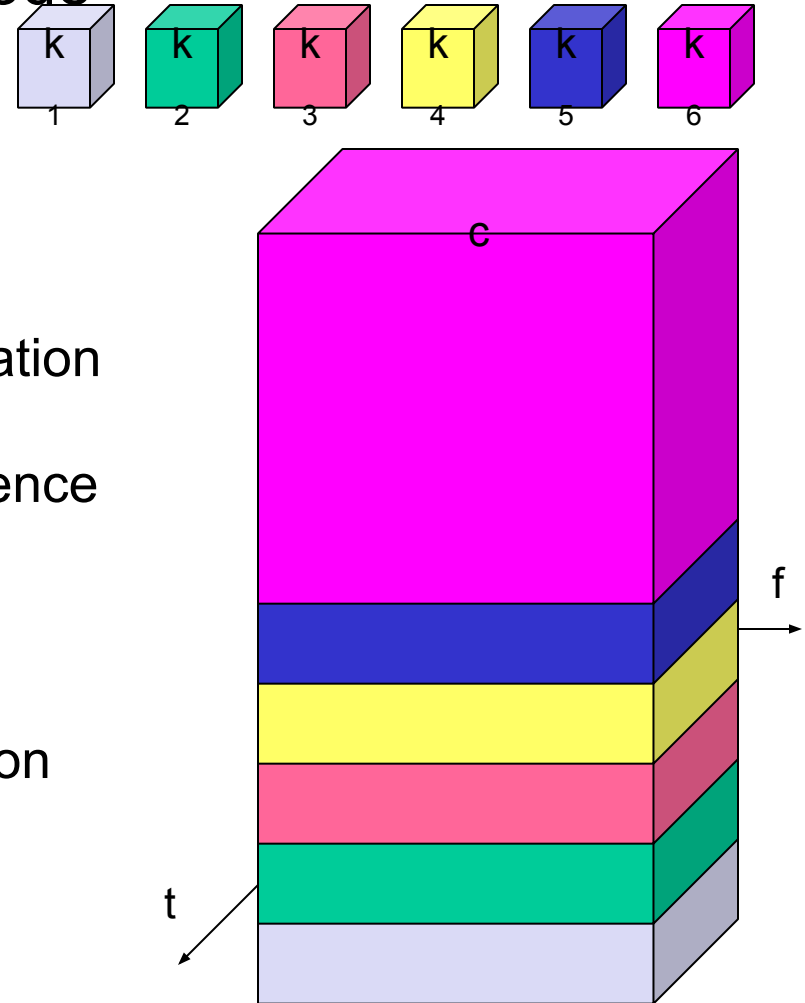
Time and frequency multiplex

- Combination of both methods
- A channel gets a certain frequency band for a certain amount of time
- Example: GSM
- Advantages:
 - better protection against tapping
 - protection against frequency selective interference
 - higher data rates compared to code multiplex
- but: precise coordination required



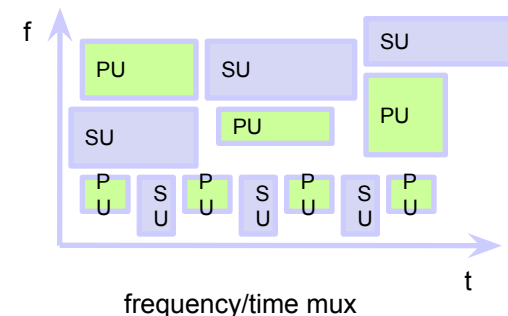
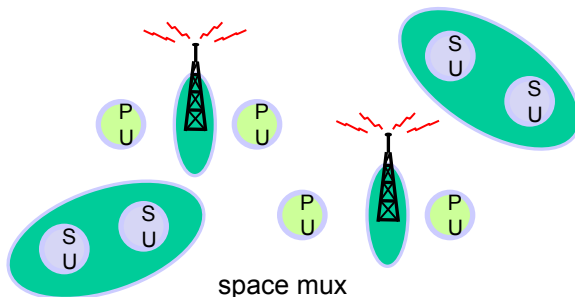
Code multiplex

- Each channel has a unique code
- All channels use the same spectrum at the same time
- Advantages:
 - bandwidth efficient
 - no coordination and synchronization necessary
 - good protection against interference and tapping
- Disadvantages:
 - lower user data rates
 - more complex signal regeneration
- Implemented using spread spectrum technology



Cognitive Radio

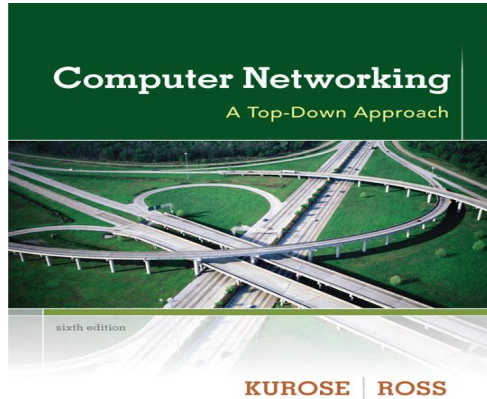
- Typically in the form of a spectrum sensing CR
 - Detect unused spectrum and share with others avoiding interference
 - Choose automatically best available spectrum (intelligent form of time/frequency/space multiplexing)
- Distinguish
 - Primary Users (PU): users assigned to a specific spectrum by e.g. regulation
 - Secondary Users (SU): users with a CR to use unused spectrum
- Examples
 - Reuse of (regionally) unused analog TV spectrum (aka white space)
 - Temporary reuse of unused spectrum e.g. of pagers, amateur radio etc.



Modulation vs Multiplexing

Aspect	Modulation	Multiplexing
Definition	The process of altering a carrier signal based on the information signal.	The process of combining multiple signals for transmission over a single medium.
Purpose	To adapt a baseband signal for efficient transmission over wireless or wired channels.	To efficiently share a communication channel among multiple signals or users.
Focus	Signal transformation	Signal sharing
Examples	AM, FM, QAM, PSK	TDM (Time Division), FDM (Frequency Division), CDM (Code Division)

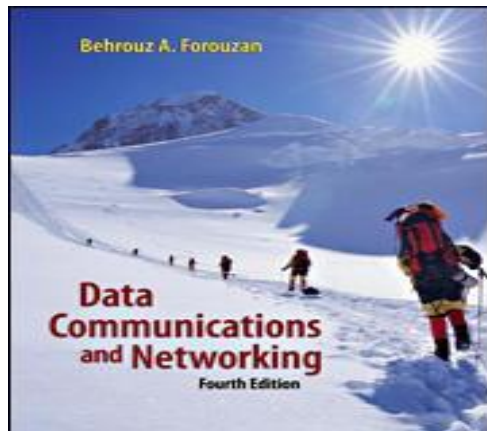
Reference



Chapter 6

Computer Networking: A Top Down Approach

by Jim Kurose, Keith Ross



Chapter 3,5, and 6

Data Communications and Networking

by Behrouz A. Forouzan

Electromagnetic Signal

Electromagnetic Signal

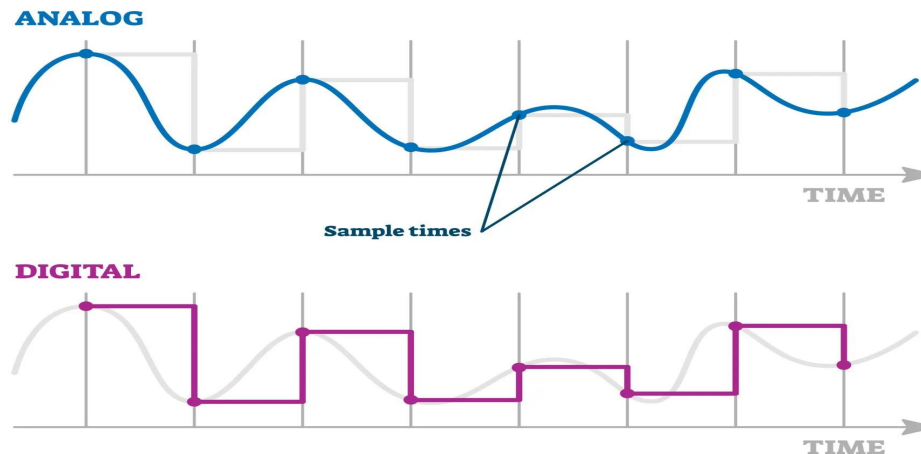
- Function of time
- Can also be expressed as a function of frequency
 - Signal consists of components of different frequencies

Electromagnetic Signal::

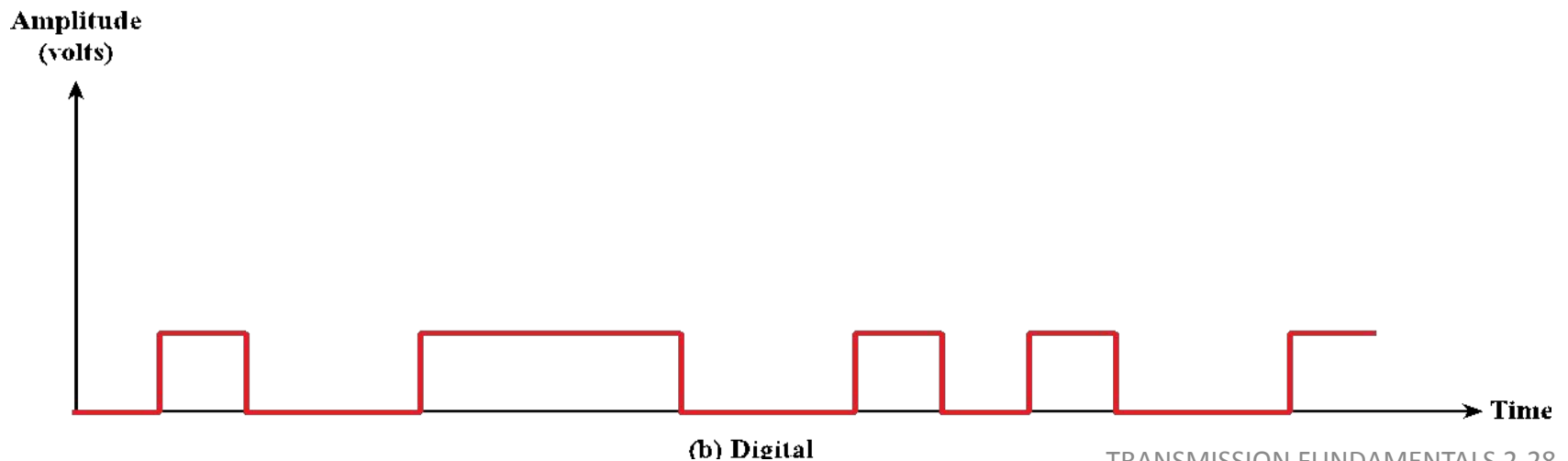
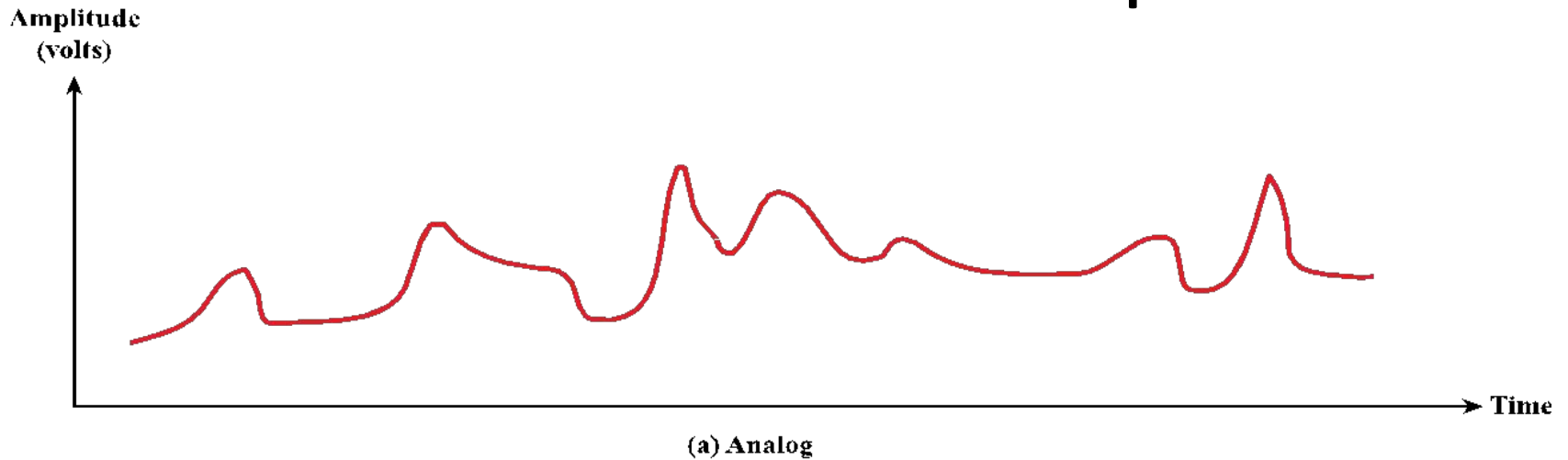
Time-Domain Concepts

- **Analog signal** - signal intensity varies in a smooth fashion over time
 - No breaks or discontinuities in the signal
- **Digital signal** - signal intensity maintains a constant level for some period of time and then changes to another constant level

ANALOG VS DIGITAL SIGNAL



Electromagnetic Signal:: Time-Domain Concepts



Electromagnetic Signal::

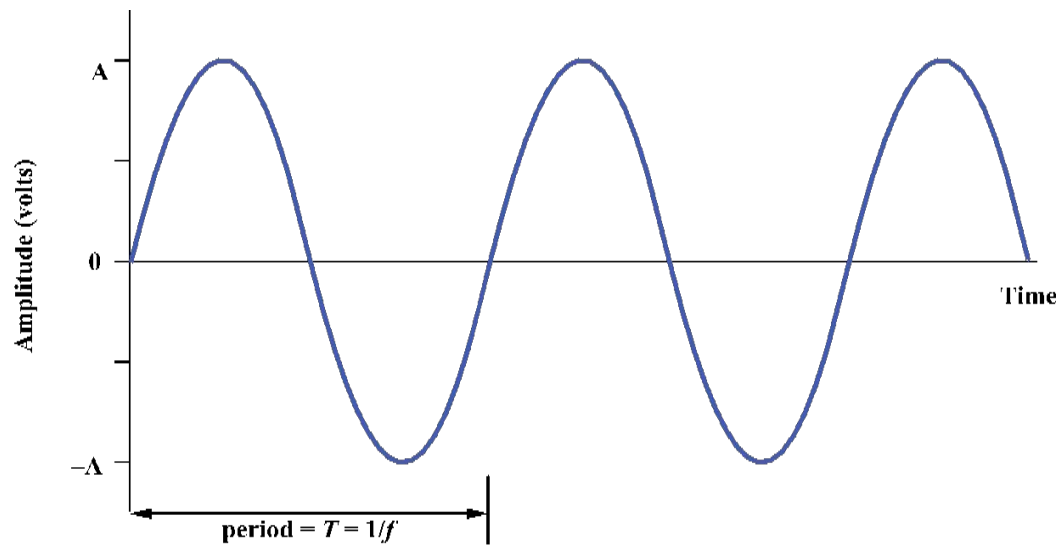
Time-Domain Concepts

- **Aperiodic signal** - analog or digital signal pattern that doesn't repeat over time
- **Peak amplitude (A)** - maximum value or strength of the signal over time; typically measured in volts
- **Frequency (f)**
 - Rate, in cycles per second, or Hertz (Hz) at which the signal repeats

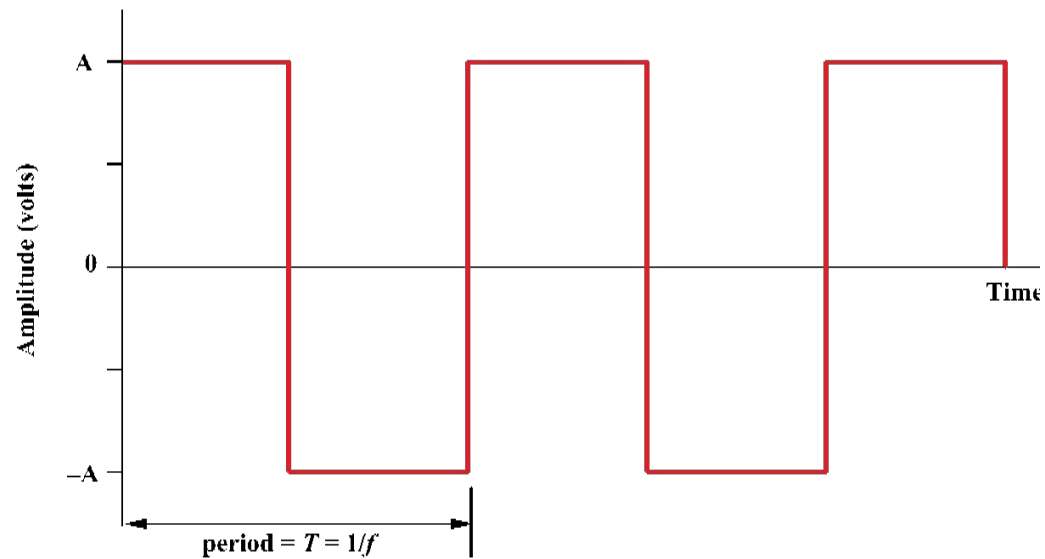
Electromagnetic Signal::

Time-Domain Concepts

- **Period (T)** - amount of time it takes for one repetition of the signal
 - $T = 1/f$
- **Phase (ϕ)** - measure of the relative position in time within a single period of a signal
- **Wavelength (λ)** - distance occupied by a single cycle of the signal
 - Or, the distance between two points of corresponding phase of two consecutive cycles



(a) Sine wave



(b) Square wave

2.2 Examples of Periodic Signals



Electromagnetic Signal:: Time-Domain Concepts

- **Periodic signal** - analog or digital signal pattern that repeats over time

$$s(t+T) = s(t) \quad -\infty < t < +\infty$$

- where T is the period of the signal

- **Types of Periodic Signals**

- Sinusoidal Wave
- Square Wave
- Triangular Wave
- Sawtooth Wave

- **Sinusoidal Wave**

- shape in the form of sine wave as shown below.
- most common signal used in communication

$$s(t) = A \cdot \sin(2\pi ft + \phi)$$



Electromagnetic Signal::

Time-Domain Concepts

Sinusoidal Wave



Where:

- **A** = amplitude (height of the wave)
- **f** = frequency (how fast it oscillates)
- **t** = time
- **φ** = phase (starting position of the wave)

$$s(t) = A \cdot \sin(2\pi ft + \phi)$$

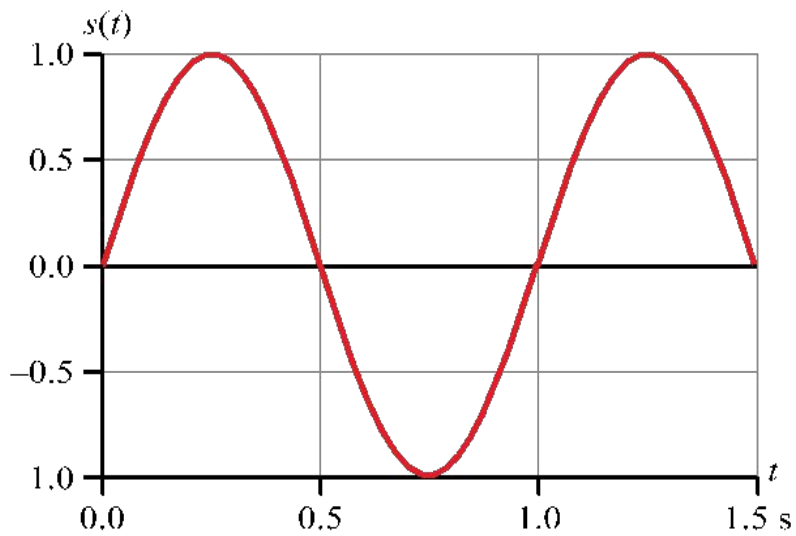
The **phase** determines **where the sine wave starts** on the time axis.

φ = **Phase shift**, measured in **radians** or **degrees**

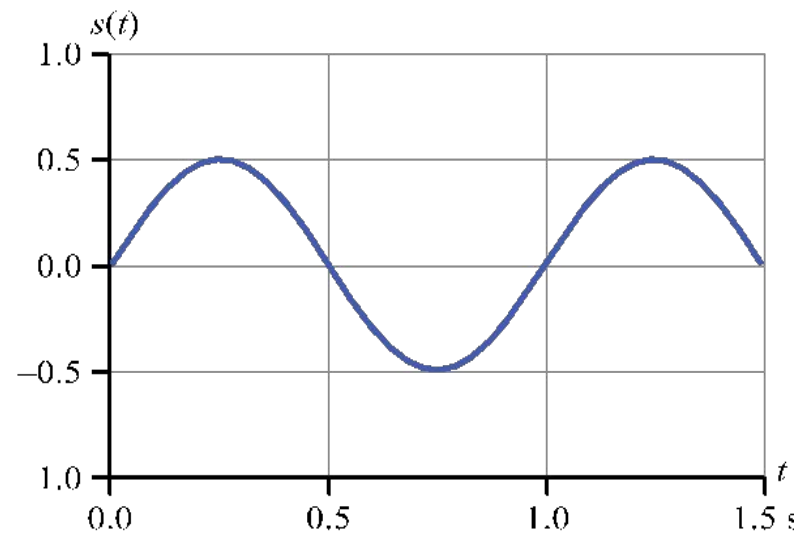
Phase Shift

What Happens

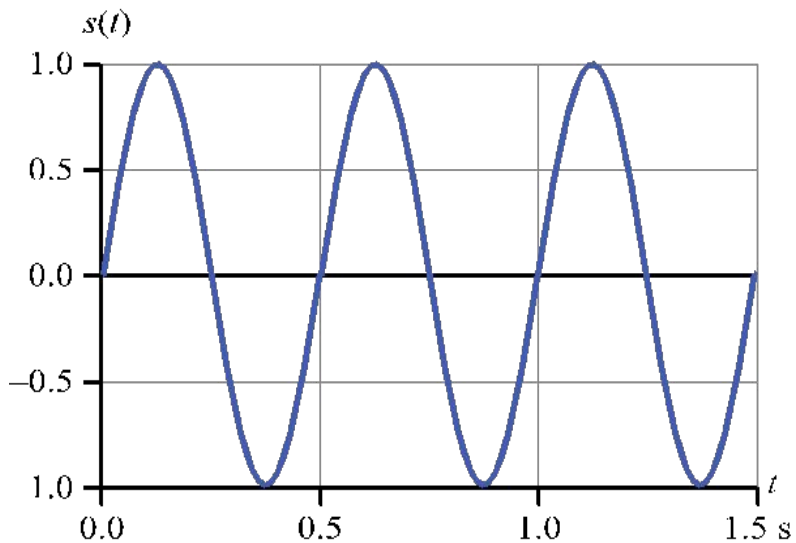
0° or 0 rad	Normal sine wave starts at 0 going up
90° or $\pi/2$ rad	Starts at the wave's peak
180° or π rad	Starts at 0 going down
270° or $3\pi/2$ rad	Starts at lowest point



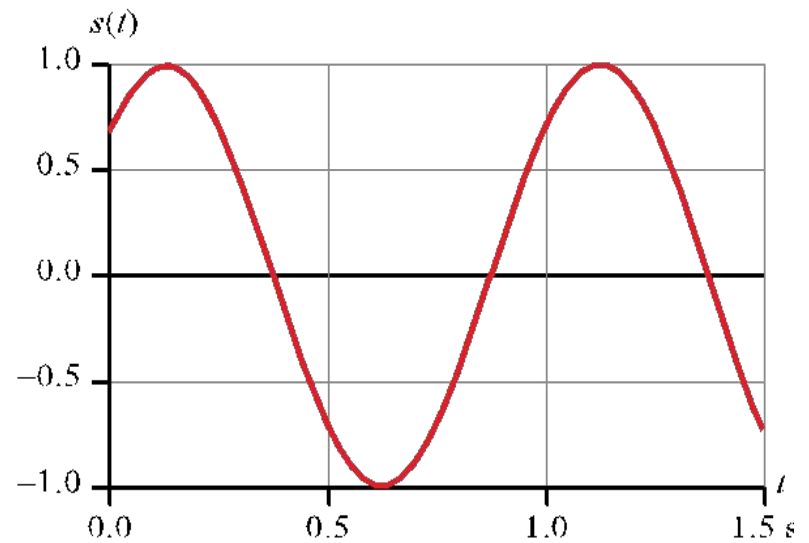
(a) $A = 1, f = 1, \phi = 0$



(b) $A = 0.5, f = 1, \phi = 0$



(c) $A = 1, f = 2, \phi = 0$



(d) $A = 1, f = 1, \phi = \pi/4$

$$2.3 \ s(t) = A \sin (2\pi ft + \phi)$$



Sine Wave Parameters

- General sine wave
 - $s(t) = A \sin(2\pi ft + \phi)$
- Figure 2.3 shows the effect of varying each of the three parameters
 - (a) $A = 1$, $f = 1$ Hz, $\phi = 0$; thus $T = 1$ s
 - (b) Reduced peak amplitude; $A=0.5$
 - (c) Increased frequency; $f = 2$, thus $T = \frac{1}{2}$
 - (d) Phase shift; $\phi = \pi/4$ radians (45 degrees)
- Note: 2π radians = $360^\circ = 1$ period

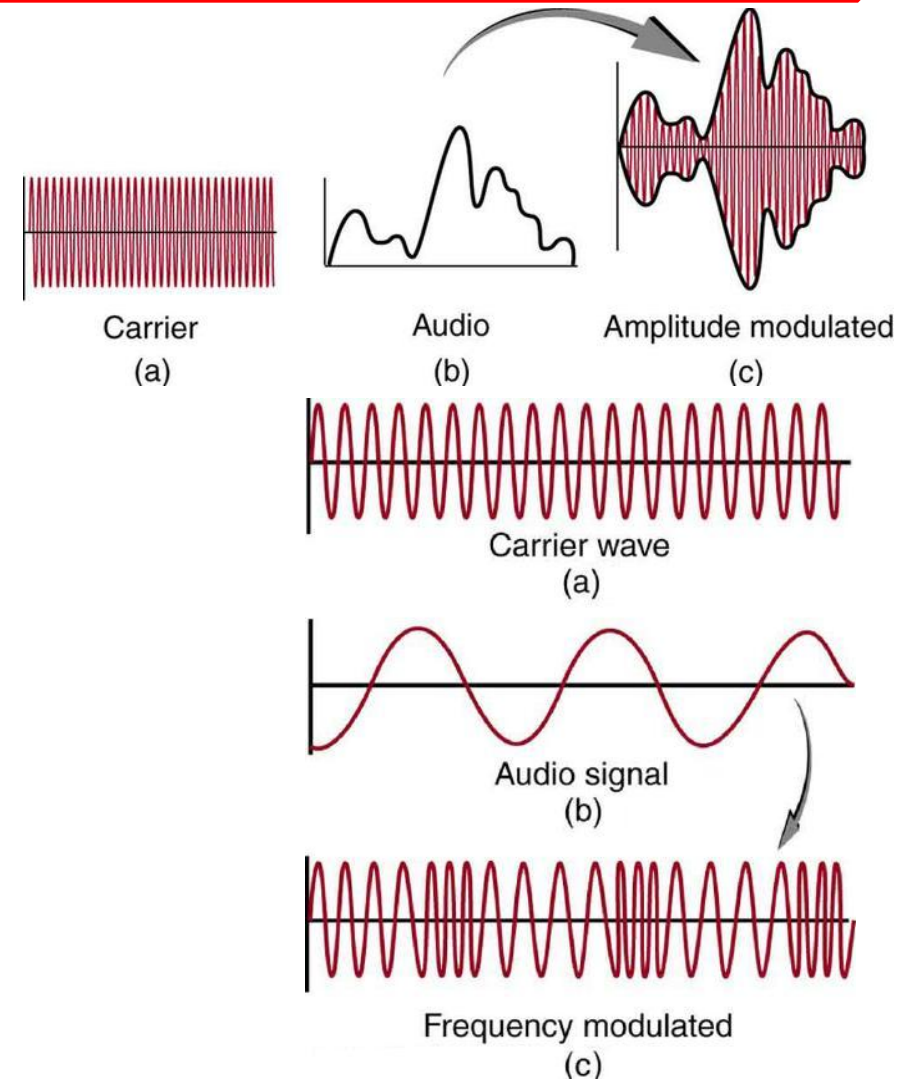
Time vs. Distance

- When the horizontal axis is *time*, as in Figure 2.3, graphs display the value of a signal at a given point in *space* as a function of *time*
- With the horizontal axis in *space*, graphs display the value of a signal at a given point in *time* as a function of *distance*
 - At a particular instant of time, the intensity of the signal varies as a function of distance from the source

Time-Domain Concepts

In Communication:

- A **carrier signal** is usually a sine wave.
- We **modulate** that sine wave by changing:
 - **Amplitude** → Amplitude Modulation (AM)
 - **Frequency** → Frequency Modulation (FM)
 - **Phase** → Phase Modulation (PM)
- OFDM uses **multiple sine waves (subcarriers)**, each orthogonal to the others.



Electromagnetic Signal::

Frequency-Domain Concepts

- Any electromagnetic signal can be shown to consist of a collection of periodic analog signals (sine waves) at different amplitudes, frequencies, and phases
- The period of the total signal is equal to the period of the fundamental frequency

Frequency-Domain Concepts

- **Fundamental frequency** - when **all frequency components** of a signal are **integer multiples of one frequency**, it's referred to as the fundamental frequency

Example

Suppose a signal is made of these frequency components:

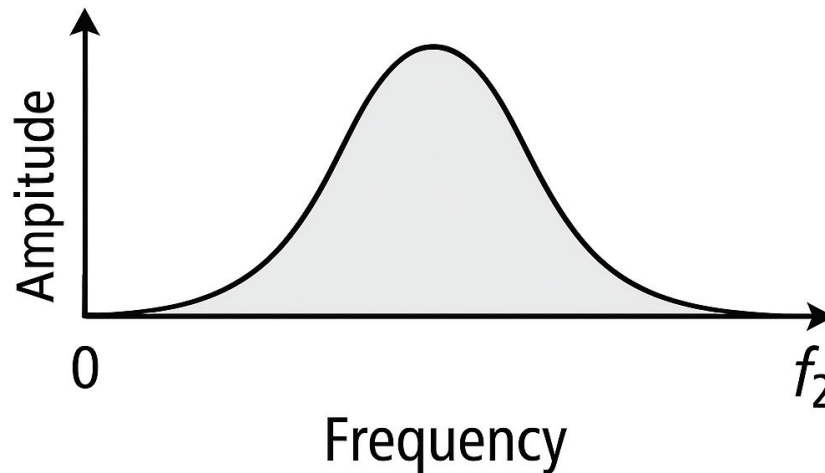
- 50 Hz
- 100 Hz
- 150 Hz
- 200 Hz
- **So, 50 Hz is the fundamental frequency**

In systems like :

- Knowing the **fundamental frequency** **helps understand the structure and periodicity** of the signal.
- It simplifies design (e.g., **subcarrier spacing in OFDM is often based on a fundamental frequency**).

Electromagnetic Signal:: Frequency-Domain Concepts

- **Spectrum** - range of frequencies that a signal contains



Electromagnetic Signal::

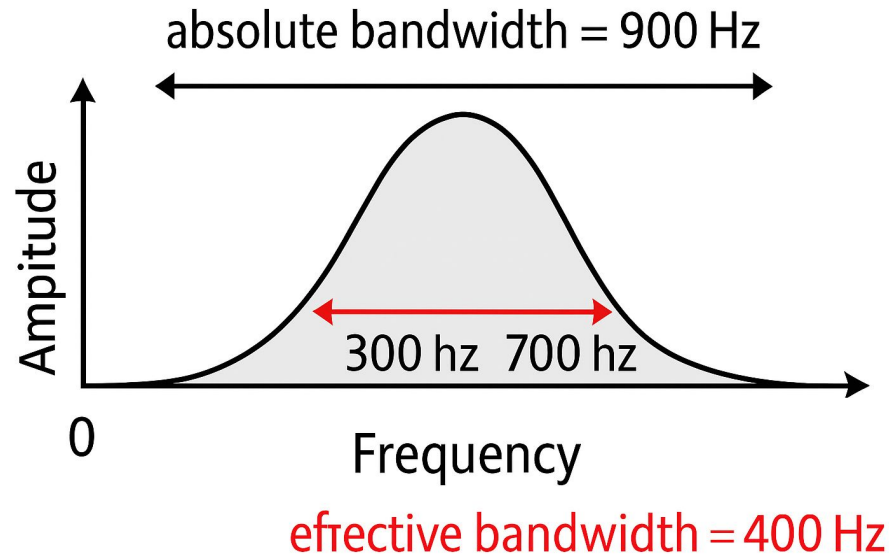
Frequency-Domain Concepts

- **Absolute bandwidth** –
 - width of the spectrum of a signal
- **Effective bandwidth** (or just bandwidth) –
 - narrow band of frequencies that most of the signal's energy is contained in
 - Focuses on where the signal is **strongest**.
- **Example:**

Suppose a signal spans:

 - From 100 Hz to 1000 Hz (absolute bandwidth = 900 Hz)
 - But 90% of the energy is between 300 Hz to 700 Hz → effective bandwidth = 400 Hz

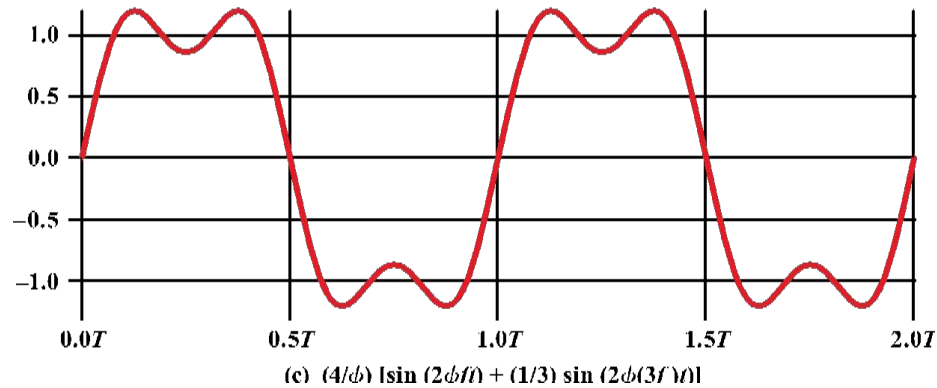
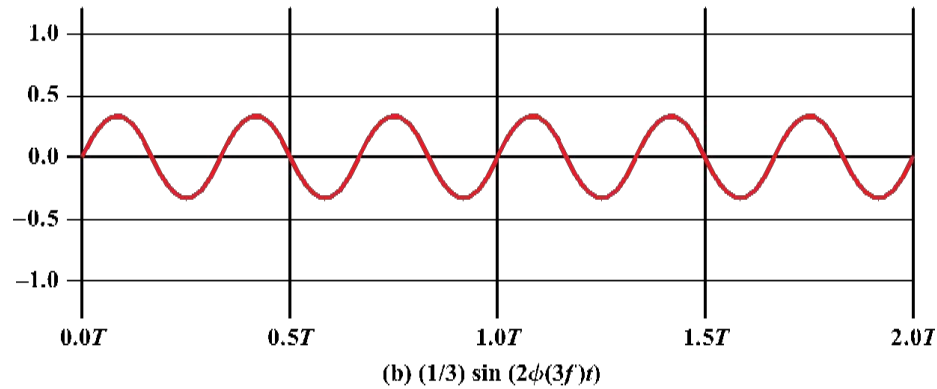
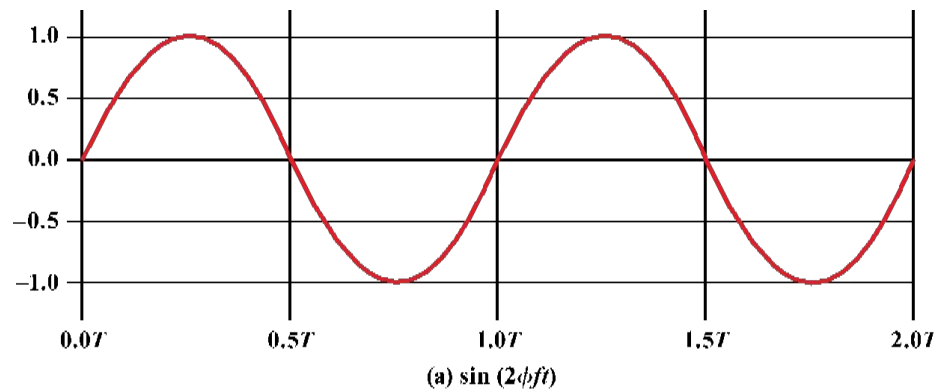
Electromagnetic Signal:: Frequency-Domain Concepts



- **Example:**

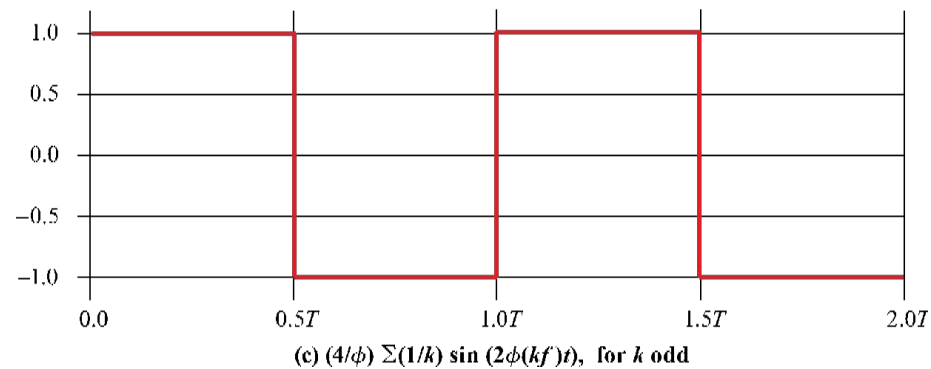
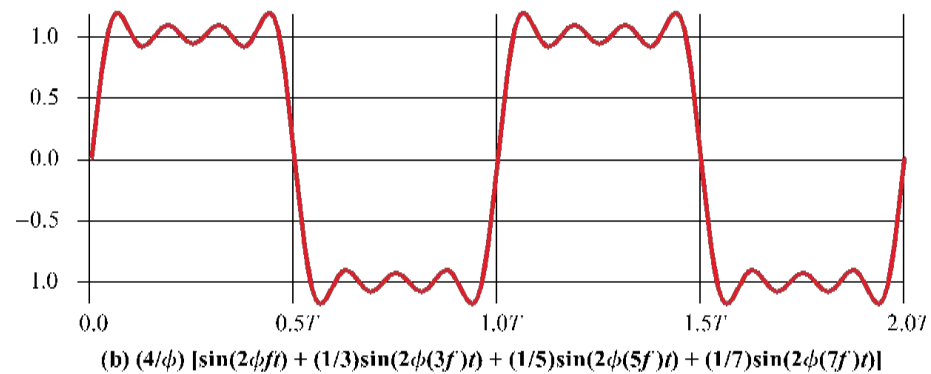
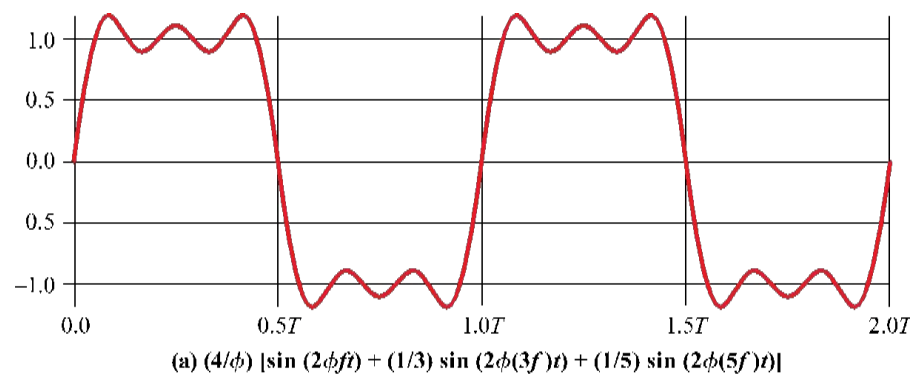
Suppose a signal spans:

- From 100 Hz to 1000 Hz (absolute bandwidth = 900 Hz)
- But 90% of the energy is between 300 Hz to 700 Hz → effective bandwidth = 400 Hz



2.4 Addition of frequency Components($T = 1/f$)





2.5 Frequency Components of Square Wave



Relationship between Data Rate and Bandwidth

- The greater the bandwidth, the higher the information-carrying capacity
- Conclusions
 - Any digital waveform will have many high-frequency components.
 - But in wireless, you can't transmit those infinite frequencies due to **antenna, RF front end**, and the **regulatory spectrum allocation**.
 - More bandwidth → higher power consumption, more **spectrum license fees**, more **complex hardware** (e.g., RFICs, ADCs)

Relationship between Data Rate and Bandwidth

- Conclusions
 - Wideband receivers are **costlier and more power-hungry** (important for phones, IoT devices).
 - in **multipath environments**, not all frequency components reach the receiver equally → **signal distortion** and errors.

Summary:

In wireless systems, **bandwidth is everything**:

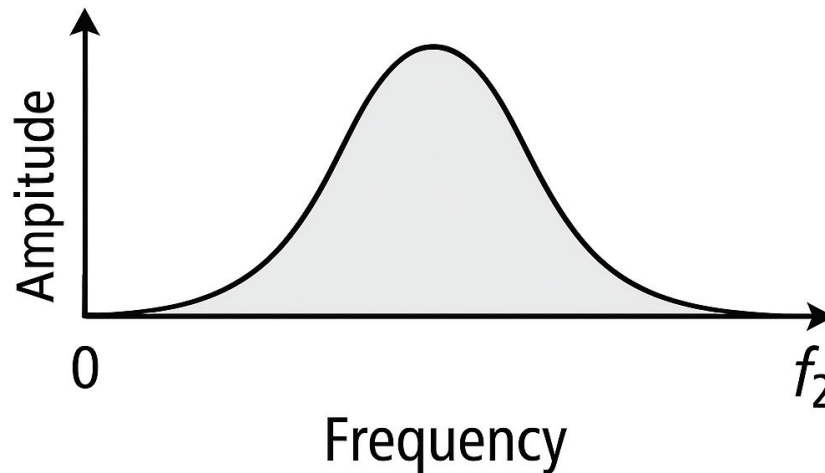
- ✓ More bandwidth → higher data rates
- ✗ But it's **costly, regulated**, and hard to handle
- ✗ Limiting it can **degrade performance** due to distortion and loss of key signal components

Technology	Bandwidth	Max Data Rate
4G LTE	~20 MHz	~100 Mbps
5G mmWave	400 MHz+	1–10 Gbps

Frequency Spectrum

Frequency Spectrum

- **Spectrum** - range of frequencies that a signal contains



Spectrum considerations

- Controlled by regulatory bodies
 - Carrier frequency
 - Signal Power
 - Multiple Access Scheme
 - Divide into time slots – Time Division Multiple Access (TDMA)
 - Divide into frequency bands – Frequency Division Multiple Access (FDMA)
 - Different signal encodings – Code Division Multiple Access (CDMA)

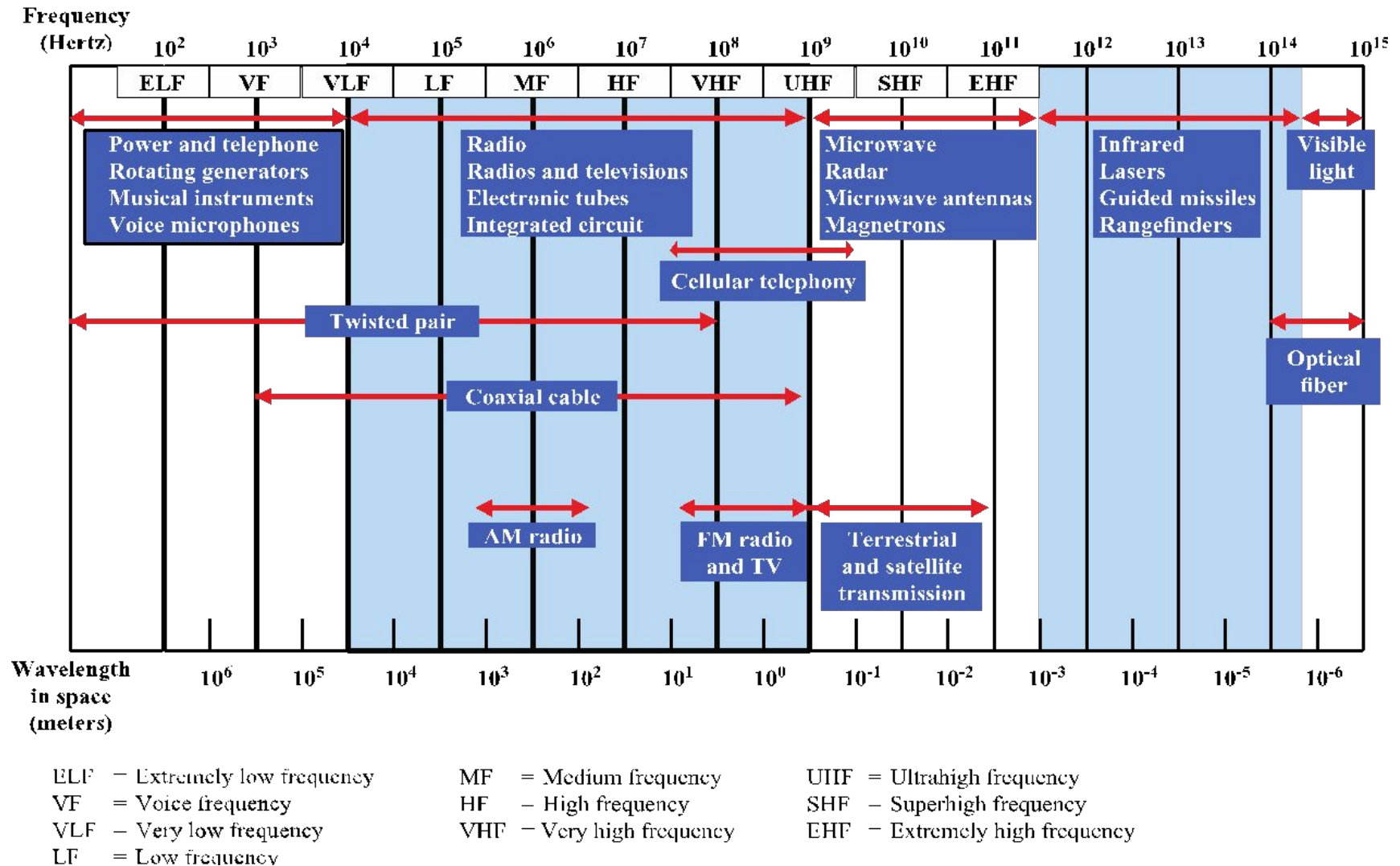
Spectrum considerations

- Federal Communications Commission (FCC) in the United States regulates spectrum
 - Military
 - Broadcasting
 - Public Safety
 - Mobile
 - Amateur
 - Government exclusive, non-government exclusive, or both
 - Many other categories

Spectrum considerations

- Industrial, Scientific, and Medical (ISM) bands
 - Can be used without a license
 - As long as power and spread spectrum regulations are followed
- ISM bands are used for
 - WLANs
 - Wireless Personal Area networks
 - Internet of Things

Frequency Spectrum



Frequency Spectrum

Frequencies for communication

VLF = Very Low Frequency

LF = Low Frequency

MF = Medium Frequency

HF = High Frequency

VHF = Very High Frequency

UHF = Ultra High Frequency

SHF = Super High Frequency

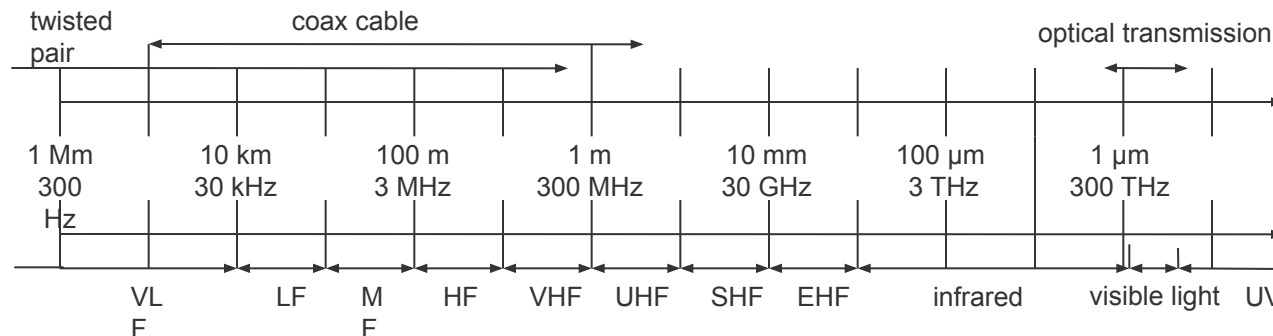
EHF = Extra High Frequency

UV = Ultraviolet Light

Frequency and wave length

- $\lambda = c/f$

- wave length λ , speed of light $c \approx 3 \times 10^8 \text{ m/s}$, frequency f



Frequency Spectrum

Example frequencies for mobile communication

VHF-/UHF-ranges for mobile radio

- simple, small antenna for cars
- deterministic propagation characteristics, reliable connections

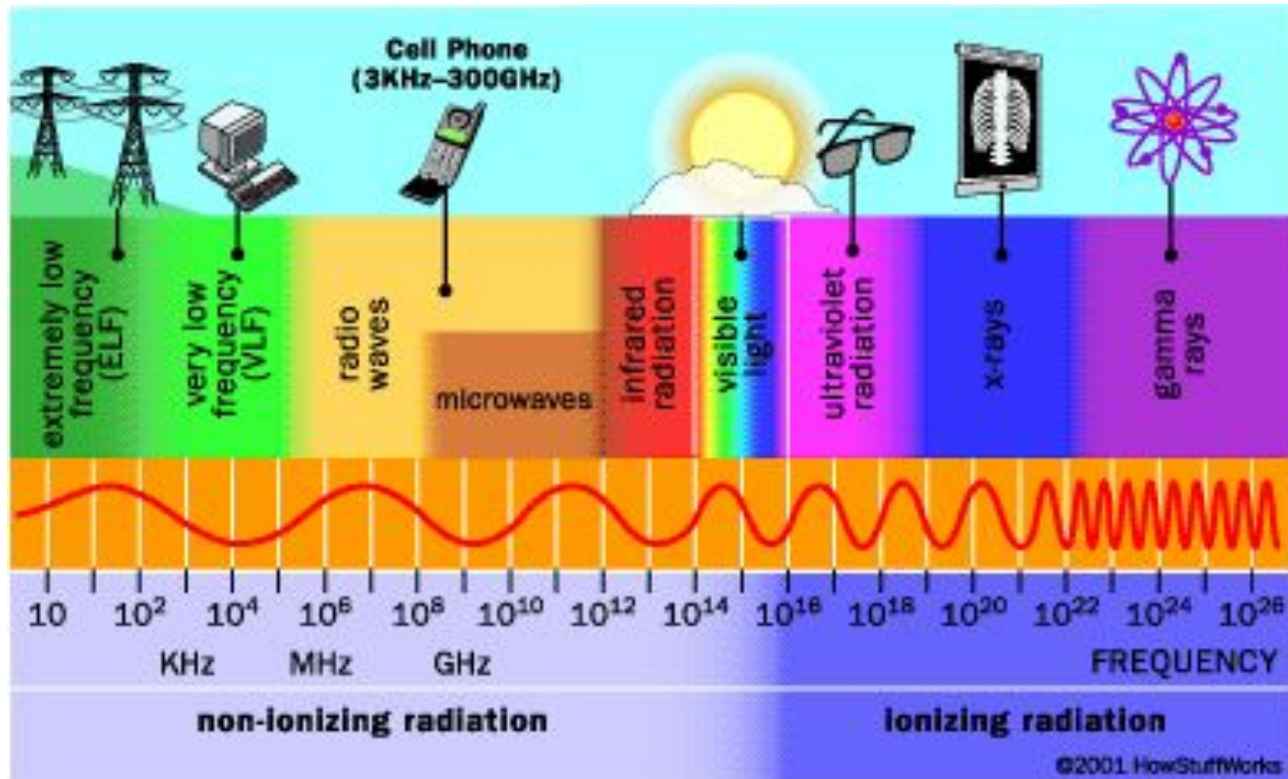
SHF and higher for directed radio links, satellite communication

- small antenna, beam forming
- large bandwidth available

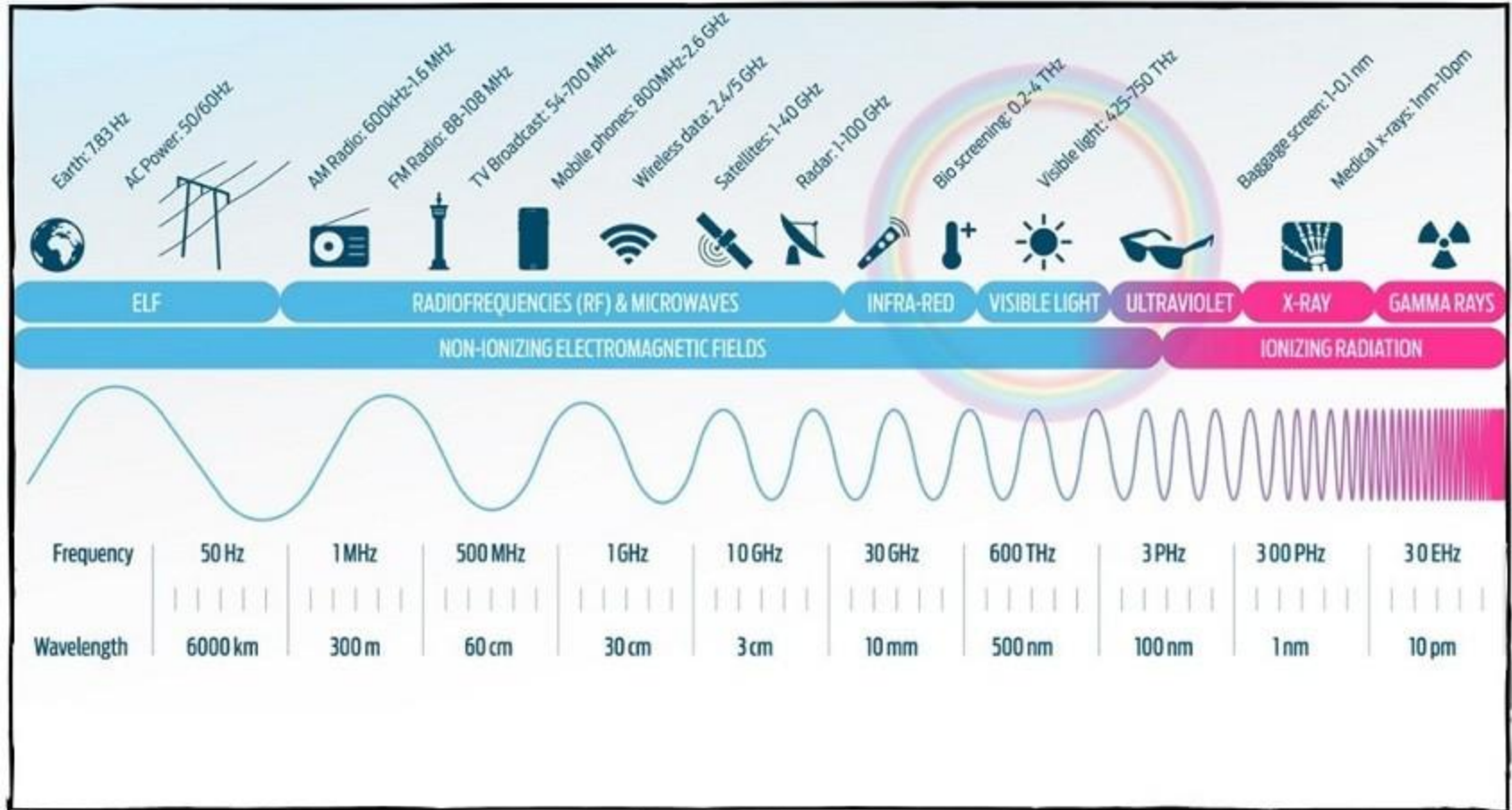
Wireless LANs use frequencies in UHF to SHF range

- some systems planned up to EHF
- limitations due to absorption by, e.g., water (dielectric heating, see microwave oven)
 - weather dependent fading, signal loss caused by heavy rainfall etc.

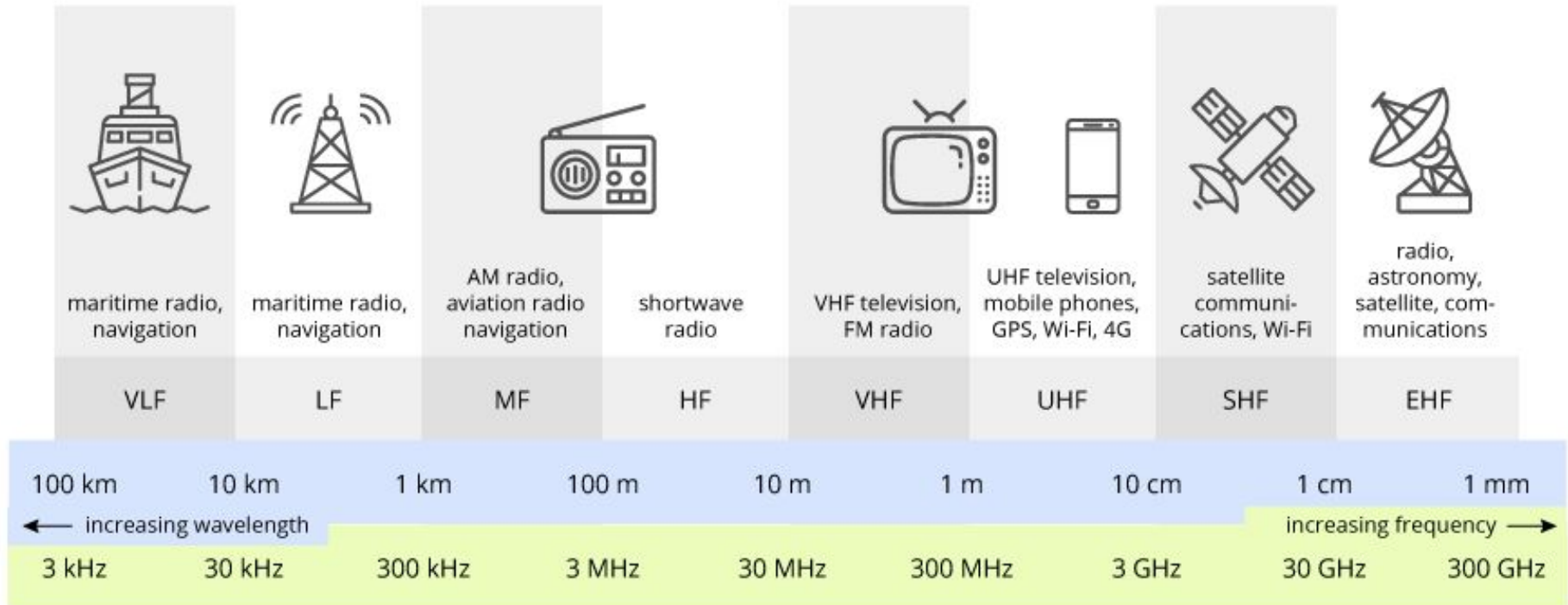
Frequency Spectrum



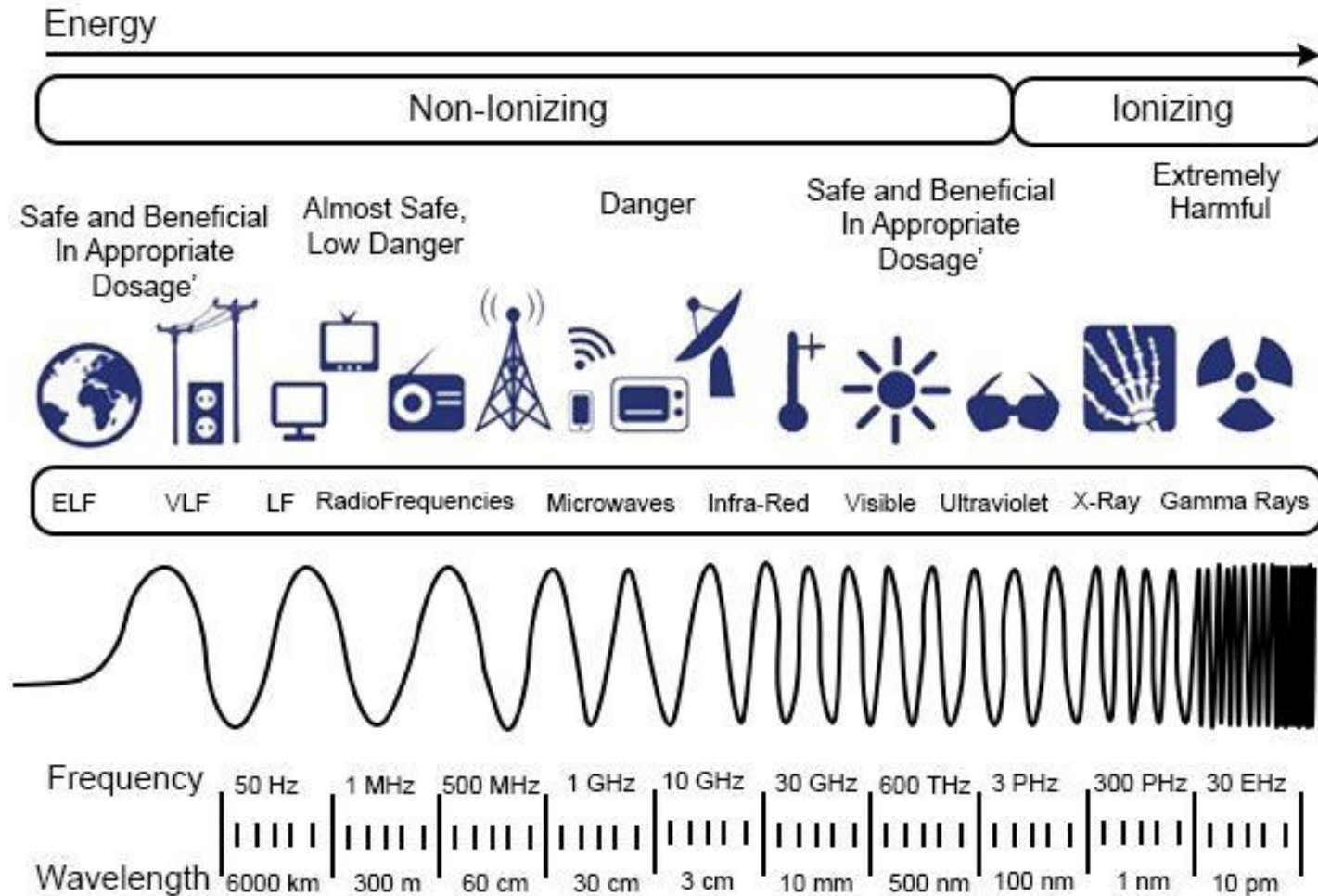
Frequency Spectrum



Frequency Spectrum



Frequency Spectrum



Combating problems:: Modulation

•Modulation

- Map the digital data onto analog signals
- Modulation is the process of changing a carrier signal to carry information (data).
- Goal: Send more bits per signal.
- In Wireless, the Carrier is a Sine Wave
- To send data, we modulate:
 - Amplitude (A) → Amplitude Shift Keying (ASK)
 - Frequency (f) → Frequency Shift Keying (FSK)
 - Phase (ϕ) → Phase Shift Keying (PSK)
 - Or combinations → QAM (Quadrature Amplitude Modulation)
- How This Works Visually: QPSK
 - Let's say we want to send 4 different symbols using QPSK:
 - We assign 00, 01, 10, 11 to four different phase angles of the carrier signal.
 - These are all sine waves of the same frequency, but with different starting points (phase shifts).

Combating problems:: Modulation

•Modulation

Visual Analogy:

- Imagine each subcarrier is a musical note .
- You can make it **louder/quieter (amplitude)** or **change the timing (phase)** to send secret messages — that's modulation!

Concept

In Modulation

Input

Digital bits (e.g., 1010...)

Carrier wave

Sine wave at a certain frequency

Output

Modified wave carrying data

Visualization

**Shift in wave's
amplitude/phase/frequency**

Combating problems:: Error Coding & Equalization

- **Error Control Coding**

- Adds **extra bits** to detect and fix errors.
- *Example:* Like adding a checksum or parity bits in your data so you can spot and fix mistakes.

- **Adaptive Modulation and Coding**

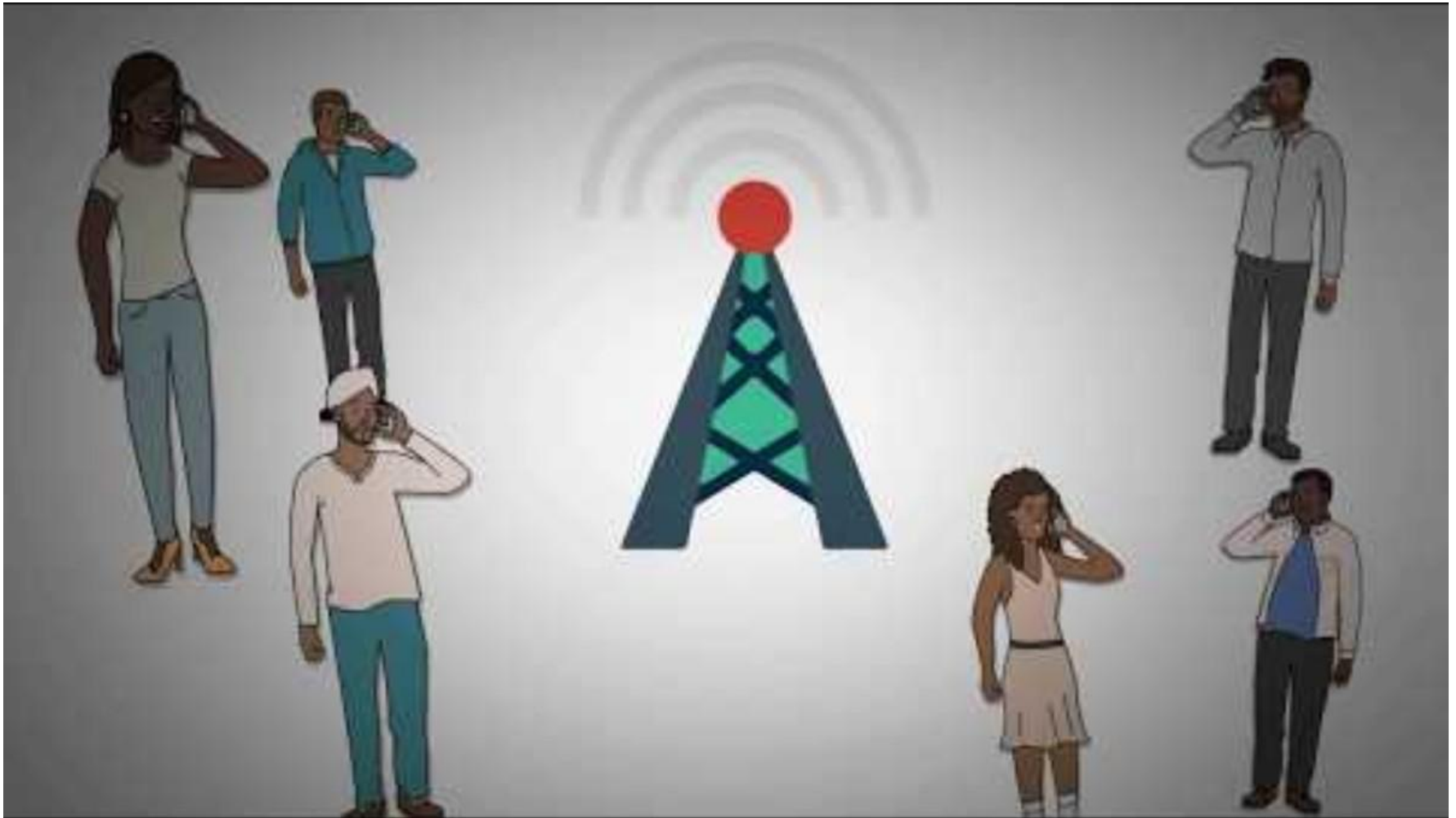
- **Dynamically adjusts** modulation/coding based on signal quality.
- *Example:* If the signal is weak, switch to safer but slower mode; if strong, go faster.

- **Equalization**

- Reduces effects of **multipath**, where signals bounce and cause distortion.
- *Example:* Like tuning your ear to separate overlapping echoes in a large hall.

Combating problems:: Multiplexing

- Multiplexing & Multiple Access



Multiplexing

- A communication medium (like a cable or channel) can usually carry **more data** than just **one signal**.
- So, we use **multiplexing** to send **multiple signals at the same time** over that medium.
- This makes **better use** of the channel and **saves resources**.



2.11 Multiplexing

Multiplexing

- Frequency-division multiplexing (FDM)
 - Takes advantage of the fact that the useful bandwidth of the medium exceeds the required bandwidth of a given signal
- Time-division multiplexing (TDM)
 - Takes advantage of the fact that the achievable bit rate of the medium exceeds the required data rate of a digital signal

Combating problems:: Multiplexing

- Multiplexing**

- What it does: Combines multiple data streams for transmission on the same physical medium.

- Types:

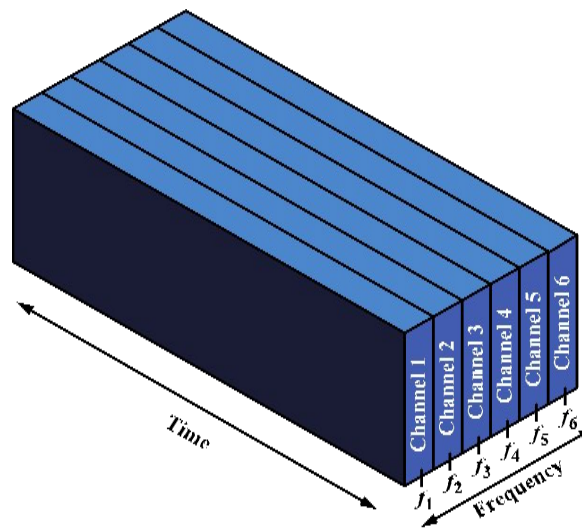
- Time Division Multiplexing (TDM)
- Frequency Division Multiplexing (FDM)
- Code Division Multiplexing (CDM)
- Space Division Multiplexing (SDM)

- Where: implemented at the physical layer (but different function than modulation).

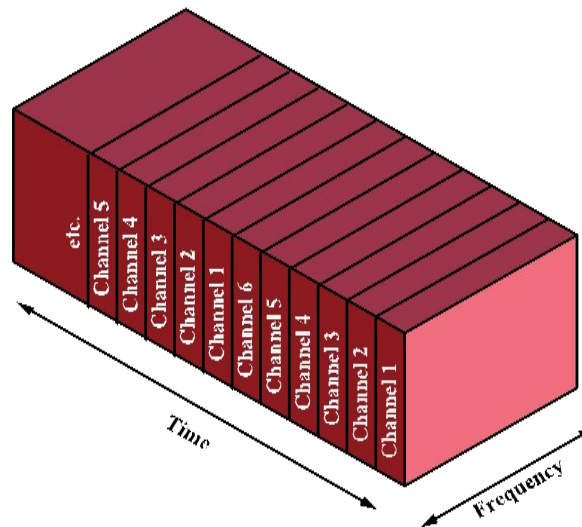
- Analogy:** Like scheduling multiple speakers to use a single microphone one after another (TDM) or letting them speak at different pitches (FDM).

- Multiple Access**

- Multiple Access refers to how multiple users share the same communication medium simultaneously — it's about who gets to transmit when and how.



(a) Frequency division multiplexing



(b) Time division multiplexing

2.12 FDM and TDM



Reasons for Widespread Use of Multiplexing

- Cost per kbps of transmission facility declines with an increase in the data rate
- Cost of transmission and receiving equipment declines with increased data rate
- Most individual data communicating devices require relatively modest data rate support